

BIOL-1 Practice Test 07

Module 11: Genomics and Biotechnology

Instructions: This practice test covers material from Module 11. Answer all questions to the best of your ability.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

1. A restriction enzyme:

- A) Copies DNA
- B) Cuts DNA at a specific sequence
- C) Joins DNA fragments
- D) Translates mRNA

2. "Sticky ends" produced by restriction enzymes are useful because:

- A) They can base-pair with matching sticky ends from another fragment
- B) They destroy unwanted DNA
- C) They prevent mutations
- D) They are found on all chromosomes

3. DNA ligase:

- A) Cuts DNA
- B) Copies DNA
- C) Joins (seals) DNA fragments together
- D) Separates DNA strands

4. A plasmid is:

- A) A human chromosome
- B) A small circular DNA molecule found in bacteria
- C) A type of RNA
- D) A protein that cuts DNA

5. Recombinant DNA is:

- A) DNA that has been destroyed
- B) DNA made from two different sources
- C) DNA that has been methylated
- D) Normal bacterial DNA

6. In PCR, what happens during the denaturation step (~95°C)?

- A) Primers bind to DNA
- B) DNA strands separate
- C) New DNA is synthesized
- D) DNA is cut by restriction enzymes

7. In PCR, what happens during the annealing step (~55°C)?

- A) DNA strands separate
- B) Primers bind to the target DNA
- C) New DNA strands are built
- D) The DNA is destroyed

8. Why is Taq polymerase used in PCR instead of normal DNA polymerase?

- A) It is faster
- B) It can survive the high heat of the denaturation step
- C) It does not need primers
- D) It works at room temperature

9. Starting with 1 DNA molecule, how many copies do you have after 5 PCR cycles?

- A) 5
- B) 10

- C) 16
- D) 32

10. In gel electrophoresis, DNA migrates toward the:

- A) Negative electrode
- B) Positive electrode
- C) Center of the gel
- D) Top of the gel

11. In gel electrophoresis, smaller DNA fragments:

- A) Stay near the top (wells)
- B) Travel farther from the wells
- C) Don't move at all
- D) Move toward the negative end

12. A DNA ladder in gel electrophoresis is used as:

- A) A source of restriction enzymes
- B) A size reference (known fragment sizes)
- C) A dye to see the DNA
- D) A primer for PCR

13. STRs (Short Tandem Repeats) are used in:

- A) PCR
- B) DNA fingerprinting (forensics and paternity tests)
- C) Transcription
- D) Translation

14. In a paternity test, which bands in the child must come from the biological father?

- A) All bands
- B) The bands that don't match the mother
- C) The largest bands
- D) The smallest bands

15. A transgenic organism (GMO) is an organism that:

- A) Has had a gene deleted
- B) Contains a gene from another species
- C) Has been exposed to radiation
- D) Is a clone

16. To make bacteria produce human insulin, scientists:

- A) Feed bacteria insulin
- B) Insert the human insulin gene into a bacterial plasmid
- C) Remove all bacterial genes
- D) Inject insulin into bacteria

17. Gene therapy treats disease by:

- A) Removing all of a patient's DNA
- B) Introducing a functional copy of a defective gene
- C) Replacing all cells in the body
- D) Changing the patient's diet

18. In ex vivo gene therapy:

- A) Treatment is delivered directly into the body
- B) Cells are removed, modified in the lab, and returned to the patient
- C) The patient's DNA is entirely replaced
- D) CRISPR is never used

19. CRISPR-Cas9 is:

- A) A type of restriction enzyme
- B) A precise gene-editing tool that can cut DNA at a specific location
- C) A method of gel electrophoresis
- D) A virus used in gene therapy

20. The difference between somatic gene therapy and germline modification is:

- A) Somatic therapy changes gametes; germline changes body cells
- B) Somatic therapy changes body cells only; germline changes are inherited
- C) They are the same thing
- D) Germline is safer

21. Golden Rice is an example of:

- A) Natural selection
- B) A transgenic crop designed to provide Vitamin A
- C) A pesticide
- D) Selective breeding

22. Why can bacteria read a human gene and make a human protein?

- A) Bacteria and humans have the same genes
- B) The genetic code is universal (same codons, same amino acids)
- C) Bacteria can mutate their DNA
- D) Human genes are simpler than bacterial genes

23. The Human Genome Project:

- A) Created the first transgenic organism
- B) Mapped all ~3 billion base pairs of human DNA
- C) Discovered restriction enzymes
- D) Invented PCR

24. One ethical concern about germline editing is:

- A) It only works in bacteria
- B) Changes are inherited by future generations
- C) It cannot fix genetic diseases
- D) It is too slow

25. Bioinformatics is:

- A) Using chemicals to modify DNA
- B) Using computational tools to analyze biological data

- C) A type of gene therapy
 - D) A form of natural selection
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Part B: Fill in the Blank (10 questions)

- 26. Restriction enzymes cut DNA at specific _____ sequences.
 - 27. The enzyme that joins DNA fragments together is _____.
 - 28. PCR stands for _____.
 - 29. PCR uses a heat-stable enzyme called _____ polymerase.
 - 30. In gel electrophoresis, DNA moves toward the positive electrode because DNA is _____ charged.
 - 31. The highly variable DNA regions used for forensic identification are called _____ (abbreviation).
 - 32. An organism that contains a gene from another species is called a _____ organism.
 - 33. Introducing a working copy of a defective gene to treat disease is called _____.
 - 34. CRISPR-Cas9 works by using a _____ RNA to guide the Cas9 protein to the target DNA sequence.
 - 35. The genetic code is nearly _____, meaning the same codons code for the same amino acids across species.
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Part C: Short Answer (4 questions)

- 36. Describe the three steps of one PCR cycle. What happens at each temperature?

37. Explain the basic steps of creating a GMO bacterium that produces human insulin. Use these terms: restriction enzyme, plasmid, DNA ligase, transformation.

38. How does gel electrophoresis separate DNA fragments? Which fragments travel farthest?

39. What is the difference between somatic gene therapy and germline modification? Why is germline modification considered more ethically controversial?